

Residual-Basic Thermodynamics of Static Bach-Flat Black Holes

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Companion-paper scaffold, 24 July 2026

Abstract

This companion isolates the static material removed from the Lorentzian endpoint-nonselection paper. It treats the local Laurent classification of static spherically symmetric Bach-flat metrics, the corrected Einstein sheet, the residual-basic normalization of the horizon generator, and simultaneous first laws on multiple horizons. The separation prevents static normalization questions from obscuring the dynamical perturbation theorem.

1 Scope

The source material for this draft is the static sector of the former unified pure-Weyl manuscript. Its claims are independent of the axial scattering and QNM computations. The final paper will retain:

1. the complete local Laurent branch classification;
2. the corrected identification of the Einstein sheet;
3. the quotient-basic horizon generator;
4. the simultaneous horizon first laws;
5. the residual normalization freedom in the absence of a preferred asymptotic clock.

2 Central normalization issue

For a static Killing generator $\xi = \partial_t$, replacement by

$$\xi \longmapsto f(J)\xi$$

with J a residual invariant changes surface gravity and Hamiltonian normalization together. A thermodynamic statement is therefore basic on the residual quotient only after its generator prescription is declared. The final companion will formulate all first-law identities in that quotient language and will not use the perturbative endpoint problem to fix an otherwise absent asymptotic clock.

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3 Separation from the radiation paper

The static classification supports the broader pure-Weyl programme, but it is not needed for the theorem that ordinary Lorentzian black-hole endpoint conditions fail to select the Einstein perturbation sector. It is therefore developed here as a logically independent paper.